More on Pivotal Quantities.

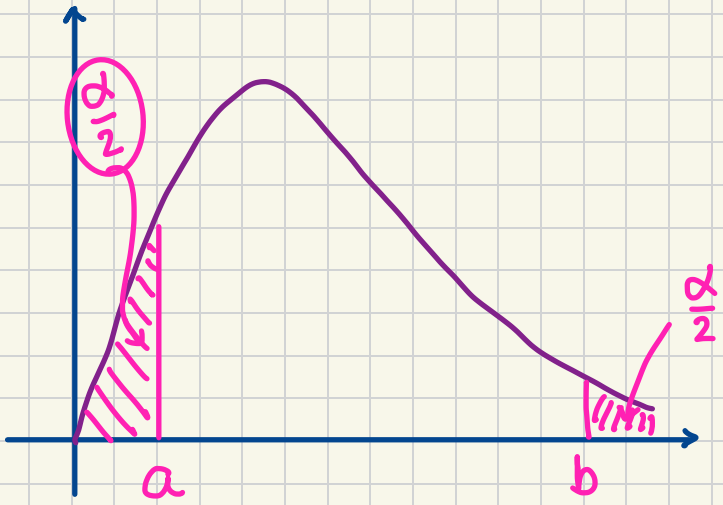
Example. $Y_1, \dots, Y_n \sim E(\tau)$ w/ τ unknown

$$\bar{Y} = \frac{1}{n}(Y_1 + \cdots + Y_n)$$

$Y_1 + \cdots + Y_n \sim \Gamma(k = n, \text{scale} = \tau)$

$$U = \frac{1}{\tau}\cdot\bar{Y} = \frac{1}{\tau}\cdot\frac{1}{n}(Y_1 + \cdots + Y_n) \sim \Gamma\left(n, \frac{\tau}{n\cdot\tau}\right) = \Gamma\left(n, \frac{1}{n}\right)$$

The dist'n doesn't depend on τ, so we have a PIVOTAL QUANTITY.



Having picked a confidence level $C = 0.9$, i.e., $\alpha = 0.10$

$a = $ qgamma(0.05, shape = n, scale = 1/n) ←

$b = $ qgamma(0.95, shap = n, scale = 1/n)

We know that

$$\mathbb{P}[a \le U \le b] = 1 - 0.10 = 0.9 = C$$

$$\mathbb{P}\left[a \le \frac{1}{\tau}\cdot\bar{Y} \le b\right] = 0.90$$

$$\mathbb{P}\left[\frac{a}{\bar{Y}} \le \frac{1}{\tau} \le \frac{b}{\bar{Y}}\right] = 0.90$$

$$\mathbb{P}\left[\boxed{\frac{\bar{Y}}{b}} \le \tau \le \boxed{\frac{\bar{Y}}{a}}\right] = 0.90$$

$\hat{\Theta}_L$ $\hat{\Theta}_R$

Approximate Confidence Interval for p.

Consider a population in which a specific trait occurs w/ an unknown probability p.

Let $(Y_1, \dots, Y_n)$ be a random sample from the Bernoulli dist'n w/ the unknown parameter p.

Goal: Designing a confidence interval for p.

Idea: Look @ the natural point estimator for p, i.e.,

$$\bar{Y} = \frac{1}{n}\underbrace{(Y_1 + \dots + Y_n)}_{B(n,p)}$$

Note: $\boxed{\mathbb{E}[\bar{Y}] = p}$ $\Rightarrow$ the dist'n depends on p

We want a pivotal quantity.

By the de-Moivre Laplace theorem:

$$S_n := Y_1 + \dots + Y_n \sim B(n,p)$$

Then, $$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{\mathcal{D}} N(0,1)$$

Note: w/ $\bar{Y}_n = \frac{1}{n} S_n$

$$U \to \boxed{\frac{\bar{Y} - p}{\sqrt{\frac{p(1-p)}{n}}}} \xrightarrow{\mathcal{D}} N(0,1)$$

But we can create an approximate confidence interval for p!

Say, C is our confidence level.

Let $z^* = \Phi^{-1}\left(\underbrace{1-\frac{\alpha}{2}}_{\frac{1+C}{2}}\right) = \text{qnorm}\left((1+C)/2\right)$

$$\mathbb{P}\left[-z^* \leq U \leq z^*\right] \approx C$$

$$\mathbb{P}\left[-z^* \leq \frac{\bar{Y}-p}{\sqrt{\frac{p(1-p)}{n}}} \leq z^*\right] \approx C$$

$$\mathbb{P}\left[-z^* \cdot \sqrt{\frac{p(1-p)}{n}} \leq \bar{Y} - p \leq z^*\right] \approx C$$

$$\mathbb{P}\left[\underbrace{\bar{Y} - z^* \cdot \sqrt{\frac{p(1-p)}{n}}}_{? \ \hat{\theta}_L} \leq p \leq \underbrace{\bar{Y} + z^* \sqrt{\frac{p(1-p)}{n}}}_{? \ \hat{\theta}_R}\right] \approx C \quad \text{X} \ \ddot{\frown}$$

Usually, we write $\hat{p} = \bar{Y}$ and remember that $\hat{p}$ is a point estimator for p.

We construct our approximate confidence interval for p as

$$p = \hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Problem 16.2. Gallup's inaugural measure of global loneliness shows over one in five people worldwide (23%) said they felt loneliness "a lot of the day yesterday."[1] *However, there were considerable variations between countries. For instance, out of 1000 individuals polled in Taiwan, 11% reported having felt loneliness "a lot of the day" before. What 90%-confidence would you report for the population proportion of Taiwanese who had felt lonely the day before?*

n

$\longrightarrow$:

$\hat{p} = 0.11$

$C = 0.9 \Rightarrow z^* = \Phi^{-1}\left(\frac{1+0.9}{2}\right) = \Phi^{-1}(0.95)$

$= 1.645 = \text{qnorm}(0.95)$

$$p = 0.11 \pm 1.645 \cdot \sqrt{\frac{0.11 \cdot 0.89}{1000}}$$

□

[1] https://news.gallup.com/poll/646718/people-worldwide-feel-lonely-lot.aspx

Sample Size in the Population Proportion Case.

Q: What is the necessary sample size w/ a required margin of error m?

→:

pt. estim. ± margin of error

$\hat{p}$ ± $z^* \cdot$ std error

$$z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$